RBF-FD Solver: Code Structure and Workflow

Dennis Corraliza

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Abstract

This note describes the structure of the RBF-FD (Radial Basis Function Finite Difference) PDE solver implemented in `assembly.py`, `domain.py`, `rbf.py`, `boundary.py`, `stencils.py`, and `geometry.py`. It walks through the pipeline from node generation to solved nodal values, module by module, with just enough mathematics to explain *why* each step exists. For the full derivation of the underlying RBF-FD numerics, see the companion theory note.

Contents

1 What the solver computes

Given a domain Ω (a square or a disk), a diffusion tensor A, a source term f, and boundary data, the solver produces an approximate solution u to

$$-\nabla\cdot(A\nabla u) = f \quad \text{in } \Omega,$$

with Dirichlet conditions (u prescribed) and/or Neumann conditions ($\partial u/\partial n$ prescribed) on different parts of the boundary. It does this by:

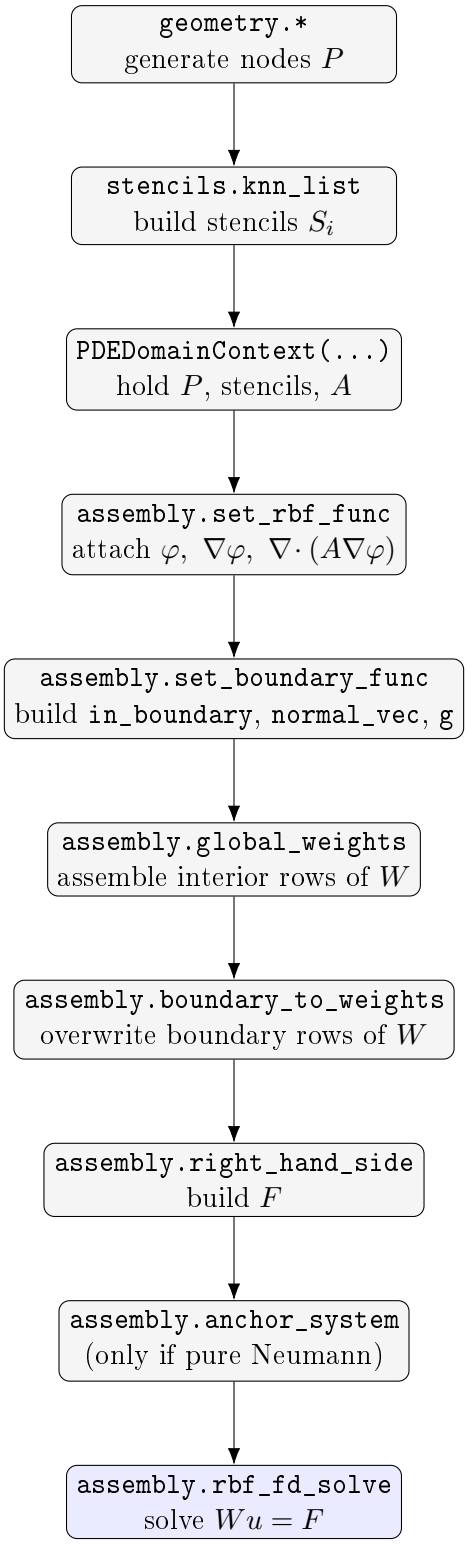
1. scattering nodes over Ω,
2. building, for each node, a small local stencil of nearby nodes,
3. fitting local differentiation weights at each node so that a weighted sum of neighboring values approximates the PDE operator there,
4. assembling all the local weights into one big sparse-like (here, dense) matrix W,
5. overwriting the boundary rows of W to enforce the boundary conditions,
6. solving the resulting linear system $Wu = F$.

The rest of this note walks through which file and function does each step.

2 Module map

Module	Responsibility
`geometry.py`	Generates node coordinates for a given domain shape.
`stencils.py`	Builds nearest-neighbor stencils for every node.
`rbf.py`	Kernel functions, their gradients, and the (anisotropic) Laplacian applied to each kernel; auxiliary grid generation.
`boundary.py`	Classifies nodes as interior/Dirichlet/Neumann; supplies boundary values and outward normals.
`domain.py`	`PDEDomainContext`: a plain data container holding everything (nodes, stencils, kernel choice, A, W, F, ...).
`assembly.py`	The actual RBF-FD machinery: local weight solves, global assembly, boundary imposition, and the top-level driver.

3 Pipeline diagram



All of these steps (except node generation and the final solve) are chained together for you by `assembly.rbf_fd_system(...)`, which is the function most user code should actually call.

4 Step 1: Node generation (`geometry.py`)

Four node-generation strategies are provided. They all return a plain array $P \in \mathbb{R}^{N\times 2}$ of node coordinates; nothing downstream cares how P was produced.

- `uniform_square(L, Nx, Ny)` – regular grid, boundary and interior points mixed together.
- `uniform_int_square(L, Nx_int, Ny_int, Nb)` – regular interior grid, plus N_b points per edge on the boundary, with corners de-duplicated. Returns `(P, num_interior)`.
- `cheby_square(L, Nx, Ny)` – Chebyshev–Gauss–Lobatto points, clustered near the edges (standard for spectral methods).
- `circular_geometry(R, Nx, Ny, num_bound)` – grid points filtered to the interior of a disk, plus points placed exactly on the circle.

```
P, num_int = geometry.uniform_int_square(L=1.0, Nx_int=40, Ny_int=40, Nb=80)
```

5 Step 2: Stencils (`stencils.py`)

For each node $\mathbf{x}_i$, `stencils.knn_list(P, k_s)` finds the *num_stencil_nodes* nearest neighbors of $\mathbf{x}_i$ (including itself) by Euclidean distance, and returns a list `stencils[i]` of their indices in P. This list is the only thing that determines which other nodal values can contribute to the discrete operator at $\mathbf{x}_i$.

6 Step 3: The domain context (`domain.py`)

`PDEDomainContext` is a plain container, not a solver — it just holds state so functions in `assembly.py` don't need ten positional arguments each. It is built once, then configured incrementally with several `set_*` calls:

```
context = domain.PDEDomainContext(P, stencils, A)
context.set_phi(...)
context.set_grad_phi(...)
context.set_laplacian_phi(...)
context.set_augmentation(True) # optional
context.set_centers(num_rings) # optional: enables least-squares weights
```

By the end of the pipeline it also holds the assembled weight matrix (`context.W`) and right-hand side (`context.F`), set via `set_W` / `set_rhs`.

7 Step 4: Choosing a kernel (`rbf.py`)

Two radial kernels are available, selected by the string `basis` passed to `assembly.set_rbf_func`:

`basis`	**Kernel** $\varphi(\mathbf{p})$	**Functions**
`'cubic'`	$\|\mathbf{p}\|^3$	`phi_cubic`, `grad_phi_cubic`, `anisotropic_diffusion_phi_cubic`
`'gaussian'`	$\exp(-(\varepsilon\|\mathbf{p}\|)^2)$	`phi_gauss`, `grad_phi_gauss`, `anisotropic_diffusion_phi_gauss`

The `anisotropic_diffusion_phi_*` functions evaluate $\nabla\cdot(A\nabla\varphi)$ in closed form (not numerically); this is what makes the RBF-FD weights exact for the kernel rather than an approximation of one. If `augmentation=True`, the linear polynomial basis $\{1, x, y\}$ (`poly_basis`, `grad_poly`, `anisotropic_diffusion_poly`) is also added, so that the local fit reproduces constants and linear functions exactly.

8 Step 5: Choosing a domain shape (`boundary.py`)

`assembly.set_boundary_func` dispatches on the string `shape` (`'square'` or `'circle'`) to build three closures:

	square	**circle**
classify node	`in_square_boundary`	`in_circular_boundary`
outward normal	`square_normal`	`disk_normal`
boundary value	`square_boundary`	`circular_boundary`

For the square, boundary data is supplied as a list of four one-variable functions `[g1, g2, g3, g4]` for the bottom, right, top, and left edges respectively (ordered counterclockwise starting at $y = 0$); for the disk, a single function `g(p)` covers the whole boundary circle.

9 Steps 6–8: Assembly (`assembly.py`)

9.1 Local weight solves

For each node, `local_weights_solve` (or, if `context.center_rings` is set, `local_weights_ls`) computes the row of differentiation weights for that node's stencil. `local_grad_solve` / `local_grad_ls` do the same for the gradient, needed only at Neumann boundary nodes. These four functions are the computational core of the method; everything else in `assembly.py` is bookkeeping around them.

9.2 Global assembly

```
W = assembly.global_weights(context, in_boundary, normal_vec)
assembly.boundary_to_weights(W, context, in_boundary, normal_vec) # in-place
F = assembly.right_hand_side(context, f, g, in_boundary)
```

`global_weights` loops over every node and scatters its local weights into a dense $N \times N$ matrix W. `boundary_to_weights` then overwrites, in place, the rows of W corresponding to Dirichlet nodes (identity row) and Neumann nodes (directional-derivative row). `right_hand_side` fills F with f at interior nodes and the boundary data at boundary nodes.

9.3 Pure-Neumann anchoring

If every boundary node is Neumann (checked by `is_pure_neumann`), W is singular (the solution is only determined up to a constant), so `anchor_system(W, F, method)` regularizes it using one of three strategies (`'mean'`, `'pin'`, or `'project'`) before solving.

10 Step 9: Solving (`assembly.rbf_fd_solve`)

```
u = assembly.rbf_fd_solve(context.W, context.F)
```

A thin wrapper around `numpy.linalg.solve`: nothing RBF-FD-specific happens here, since all of the discretization work was already baked into W and F.

11 Putting it together: the top-level driver

In practice, a user only needs to call:

```
context = assembly.rbf_fd_system(
    f, g_bound, btype, P,
    basis='cubic', shape='square', L=1.0,
    num_stencil_nodes=25, num_rings=None, # num_rings=None -> direct solve; num_rings=int
        -> least squares
    augmentation=True, A=my_tensor
)
u = assembly.rbf_fd_solve(context.W, context.F)
```

`rbf_fd_system` runs steps 3 through 8 end-to-end (printing a short progress message after each stage) and returns a fully populated `PDEDomainContext`, ready to be handed to `rbf_fd_solve`.

12 Validating the solver: manufactured solutions

To check that the solver is actually correct, test problems are built by picking a closed-form u_{exact}, then differentiating it analytically to obtain a matching f and boundary data — this is the *method of manufactured solutions*. Each such test problem is a small Python function (e.g. `example_3`) returning the tuple `(f, g, btype, u_exact)`, ready to be passed straight into `rbf_fd_system`. Once solved, `report_and_graph(context, u_exact)` compares the numerical solution against u_{exact} at every node and reports:

- the condition number of W,
- the maximum pointwise error and a discrete L^2 error,
- contour and 3D surface plots of the approximate solution, the exact solution, and the pointwise error.

This is the standard way to sanity-check a new kernel, stencil size, or boundary condition setup before trusting the solver on a problem without a known exact solution.